MA222 Review 2 Answers

1. Twelve customers at a store bought this many items each: 2, 3, 3, 4, 4, 4, 5, 5, 6, 3, 4, 14. Make a dotplot of the data. Then describe the distribution: Where is the peak? What is the range? Is there an outlier?

Answer: Tally: 2 (1 dot), 3 (3 dots), 4 (4 dots), 5 (2 dots), 6 (1 dot), 14 (1 dot). The peak is at 4. The range is $14 - 2 = 12$. The value 14 is an outlier: it is far from the rest of the data.

2. Three student clubs have these numbers of members (no student is in more than one club):

Club	Art	Chess	Dance
Members	6	8	5

a) The observational units are the clubs, and the variable is the number of members in the club. Find the mean and the median of this variable.

b) The observational units are the 19 students, and the variable is the number of members in the student's club. Find the mean and the median of this variable.

Answer: a) Mean: $\dfrac{6+8+5}{3} = \dfrac{19}{3} \approx 6.33$ members. Median: in order $5, 6, 8$, so 6 members. b) Six students report 6, eight report 8, and five report 5. Mean: $\dfrac{6 \cdot 6 + 8 \cdot 8 + 5 \cdot 5}{19} = \dfrac{36+64+25}{19} = \dfrac{125}{19} \approx 6.58$ members. Median: in order, five 5s, six 6s, then eight 8s; the middle (10th) of the 19 values is 6 members.

3. Maya and Leo take exams in different classes.

	Score	Class mean	Class std dev
Maya	88	80	6
Leo	91	85	4

Who did better compared to their own class? Use z-values to explain.

Answer: Maya: $z = \dfrac{88-80}{6} = \dfrac{8}{6} = 1.33$. Leo: $z = \dfrac{91-85}{4} = \dfrac{6}{4} = 1.5$. Leo did better compared to his class: he is 1.5 standard deviations above his class mean, and Maya is 1.33 above hers.

4. Use the standard normal table. Sketch the area, then find the left tail area for $z = -1.47$.

Answer: Row -1.4, column .07: left tail area $= 0.0708 \approx 7\%$.

5. Use the standard normal table. Sketch the area, then find the right tail area for $z = 1.21$.

Answer: Row 1.2, column .01: left tail area $= 0.8869$, so right tail area $= 1.0 - 0.8869 = 0.1131 \approx 11\%$.

6. Use the standard normal table. Sketch the area, then find the area between $z = -0.85$ and $z = 1.10$.

Answer: Left tail area for $z = 1.10$ (row 1.1, column .00) is 0.8643. Left tail area for $z = -0.85$ (row -0.8, column .05) is 0.1977. Area between $= 0.8643 - 0.1977 = 0.6666 \approx 67\%$.